

# AIM PROTOCOL

Agri • Identity • Microfinance

## Version 2.1 — Release Notes & Technical Reference

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 Post-Hackathon Development — Solana Frontier 2026

Live: <https://aim-protocol.vercel.app/>

| Status              | Live on Solana Devnet                        |
|---------------------|----------------------------------------------|
| Frontend Repo       | github.com/amjadkamara/aim-protocol          |
| Smart Contract Repo | github.com/amjadkamara/aim-program           |
| Program ID          | AhHHJTU5vodDYE2yLNet2bE6jad9F3xSfbLQdUmykKqB |
| Anchor Version      | 0.31.1                                       |
| Context             | Post-Hackathon — Solana Frontier 2026        |

### 1. Overview

Version 1.1 proved the concept — a farmer could connect a wallet, register an on-chain identity, and request a microloan on Solana devnet. V2.1 transforms that proof of concept into a production-ready foundation. Every major weakness in V1.1 has been addressed at both the smart contract and frontend level.

V2.1 is a complete overhaul of the underlying account architecture, the full loan lifecycle, the user interface, and the error handling layer. The result is a system that a real farmer can use without technical knowledge and that a real lender can trust.

## 2. What Was Wrong With V1.1

| Problem                                       | Impact                                                                              |
|-----------------------------------------------|-------------------------------------------------------------------------------------|
| Any wallet could register multiple Farmer IDs | Duplicate identities, system manipulation — no enforcement at contract level        |
| Loan accounts used random keypairs            | Frontend had to generate and store keypairs locally — fragile, insecure, unreliable |
| No loan repayment UI                          | Farmers could not close loans or re-borrow — loan cycle broken at first repayment   |
| Repaid loan PDA blocked re-borrowing          | Critical bug — program attempted to init an account that already existed            |
| No loan status visibility                     | Farmers had no dashboard to track their position, deadline, or eligibility          |
| Raw Solana errors shown to users              | custom program error: 0x0 — meaningless to farmers unfamiliar with blockchain       |
| No network or balance detection               | New users silently failed with 0 SOL or on the wrong network                        |
| No loan history                               | No borrowing record — a core requirement for any microfinance system                |

## 3. What V2.1 Built

### 3.1 PDA Enforcement — One Wallet, One Farmer ID

*Files: `programs/aim-program/src/lib.rs`*

The most important upgrade in V2.1. In V1.1, the `create_farmer_id` instruction accepted any keypair as the farmer account — nothing prevented the same wallet registering twice with a different keypair.

V2.1 derives the farmer account address deterministically from the wallet using Program Derived Addresses:

FarmerAccount PDA seeds: `["farmer", owner.key().as_ref()]`, bump

LoanAccount PDA seeds: `["loan", owner.key().as_ref()]`, bump

The blockchain itself now enforces uniqueness. A second registration attempt fails at the program level — no frontend validation required. The PDA address is deterministic: given any wallet public key, the FarmerAccount address can always be derived without storing any state.

### 3.2 Loan Account Closing on Repayment

Files: *programs/aim-program/src/lib.rs*

V1.1 marked loans as `is_repaid = true` but left the account open on-chain. When a farmer attempted to re-borrow, the program tried to initialise an account at a PDA address that already existed — returning an error and blocking the entire loan cycle permanently.

V2.1 closes the loan account on repayment using Anchor's close constraint:

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| close = owner                                                                                                   |
| Effect: Rent SOL is returned to the farmer's wallet. The PDA is freed. A new loan can be initiated immediately. |

### 3.3 close\_loan Instruction for Legacy Accounts

Files: *programs/aim-program/src/lib.rs*

A `close_loan` instruction was added to handle accounts created under V1.1 that had been marked repaid but never closed. This migration instruction allowed legacy accounts to be cleared cleanly, restoring borrow eligibility for affected wallets without data loss.

### 3.4 Comprehensive Test Suite

Files: *tests/aim-program.ts*

V1.1 had only the default Anchor boilerplate test. V2.1 ships a full 5-test suite — all tests pass before every deployment:

| Test                          | Validates                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------|
| Creates a Farmer ID via PDA   | PDA derivation correct, FarmerAccount initialised with correct fields                |
| Blocks duplicate registration | Second registration attempt from same wallet returns <code>AlreadyInUse</code> error |
| Requests a loan               | LoanAccount PDA created, amount and purpose stored                                   |

|                                         |                                                                  |
|-----------------------------------------|------------------------------------------------------------------|
|                                         | correctly                                                        |
| Blocks second loan while one is active  | ActiveLoanExists error returned correctly                        |
| Repays loan and confirms account closed | LoanAccount PDA closed, rent returned, re-borrow cycle unblocked |

### 3.5 Repay Loan UI

Files: `src/RepayLoan.jsx` (NEW)

New screen displaying full loan details with real-time status derived from on-chain data:

| Status         | Display                                                   |
|----------------|-----------------------------------------------------------|
| <b>ACTIVE</b>  | Days remaining until repayment deadline — shown in yellow |
| <b>OVERDUE</b> | Days past deadline — shown in red with urgency indicator  |
| <b>REPAID</b>  | Loan closed — farmer eligible for new loan                |

The Repay in Full button triggers the `repay_loan` instruction on-chain via Phantom. On confirmation, the loan PDA is closed and the farmer is redirected to their dashboard.

### 3.6 Loan Status Dashboard

Files: `src/Dashboard.jsx` (NEW)

Central hub for a farmer's on-chain identity. After registration, farmers land here instead of the home screen. Single screen showing:

- Farmer profile — name, crop type, district, farm size, member since date, Farmer PDA address
- Current loan status with deadline, days remaining, and status badge
- Action buttons that adapt to loan state — Request Loan if eligible, Repay if active
- Navigation to full loan history

### 3.7 Loan History

Files: `src/LoanHistory.jsx` (NEW)

On-chain transaction history parsed directly from Solana RPC signature records. No additional smart contract storage required — the blockchain's permanent transaction record serves as the history layer.

| Field Shown Per Loan  | Source                                                 |
|-----------------------|--------------------------------------------------------|
| Loan number           | Derived from transaction order                         |
| Amount (SOL)          | Parsed from transaction instruction data               |
| Status badge          | Derived from account state                             |
| Request date          | Block timestamp of loan creation transaction           |
| Repayment date        | Block timestamp of repayment transaction               |
| Solana Explorer links | Transaction signatures for both creation and repayment |

Summary statistics shown at top: total loans taken, total loans repaid, total SOL borrowed.

### 3.8 Plain English Error Messages

Files: *src/useAimProgram.js*, *src/FarmerID.jsx*, *src/Microloan.jsx*, *src/RepayLoan.jsx*

A `parseAnchorError()` helper function was added to `useAimProgram.js`. It maps every raw Solana and Anchor error code to a human-readable message displayed inline in the UI:

| Raw Error                 | Plain English Message                                                  |
|---------------------------|------------------------------------------------------------------------|
| custom program error: 0x0 | This account already exists on-chain                                   |
| ActiveLoanExists          | You already have an active loan. Repay it before requesting a new one. |
| LoanAlreadyRepaid         | This loan has already been repaid.                                     |
| AccountNotInitialized     | Account not found. Please register your Farmer ID first.               |
| insufficient funds        | Not enough SOL in your wallet to complete this transaction.            |
| User rejected             | Transaction cancelled. Please try again when ready.                    |
| BlockhashNotFound         | Network timeout. Please try again.                                     |

### 3.9 Devnet SOL & Network Warning Banner

Files: *src/WalletStatus.jsx* (NEW), *src/Layout.jsx* (UPDATED)

A persistent banner renders on every screen when a wallet is connected. It detects two conditions and warns the user before any transaction attempt fails silently:

| Condition                  | Banner                               | Action Provided                           |
|----------------------------|--------------------------------------|-------------------------------------------|
| Wrong network (not devnet) | Red — This app runs on Solana Devnet | Instructions to switch network in Phantom |
| Low SOL balance            | Yellow — Low devnet SOL detected     | Direct link to faucet.solana.com          |

Network is detected via genesis hash comparison — reliable across all Phantom versions and wallet types. Both banners are dismissable.

### 3.10 Mobile Responsive Fixes

Files: `src/App.jsx`, `src/Microloan.jsx`

- Home screen action buttons now full width on mobile
- Badge text no longer truncates on small screens
- Active loan warning banner no longer wraps awkwardly on narrow viewports
- All screens tested at 375px width (iPhone SE — minimum supported viewport)

## 4. File Changelog

| File                                         | Status  | What Changed                                                                                                             |
|----------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------|
| <code>programs/aim-program/src/lib.rs</code> | Updated | PDA seeds for farmer and loan accounts, close constraint on repayment, close_loan instruction, bump storage              |
| <code>tests/aim-program.ts</code>            | Updated | Full 5-test suite replacing default Anchor boilerplate                                                                   |
| <code>src/useAimProgram.js</code>            | Updated | PDA derivation replacing keypairs, repayLoan, closeLoan, fetchFarmer, fetchLoan functions added, parseAnchorError helper |
| <code>src/FarmerID.jsx</code>                | Updated | PDA-based registration, already-registered detection and display                                                         |
| <code>src/Microloan.jsx</code>               | Updated | PDA-based loan request, active loan detection, error messages, mobile fix                                                |
| <code>src/App.jsx</code>                     | Updated | New routes: dashboard, repayloan, loanhistory — mobile button width fix                                                  |
| <code>src/Dashboard.jsx</code>               | New     | Farmer profile card, loan status card, adaptive action buttons, loan history navigation                                  |
| <code>src/RepayLoan.jsx</code>               | New     | Loan repayment screen with real-time ACTIVE, OVERDUE, REPAYED status detection                                           |

|                          |         |                                                                            |
|--------------------------|---------|----------------------------------------------------------------------------|
| src/LoanHistory.jsx      | New     | On-chain transaction history parser with per-loan detail and summary stats |
| src/WalletStatus.jsx     | New     | Persistent network and balance warning banner                              |
| src/Layout.jsx           | Updated | WalletStatus banner integration                                            |
| src/idl/aim_program.json | Updated | V2.1 IDL reflecting PDA seeds and close_loan instruction                   |

## 5. V2.2 Roadmap

### Smart Contract

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- Credit scoring based on repayment history — stored in FarmerAccount
- Loan counter per farmer for proper history indexing
- USDC stablecoin disbursement replacing SOL — removes price volatility
- Oracle integration for crop price feeds
- Multi-sig loan approval for cooperative governance

### Frontend

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- Admin dashboard — view all registered farmers and loans live from devnet
- Public farmer profile lookup by wallet address
- KYC integration — government ID verification for donor compliance
- Push notifications for loan deadlines

### Infrastructure

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- Mainnet migration following security audit
- Custom domain via Aadios Systems
- Private production repository
- Analytics and monitoring

### Business

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- Pilot with one African farming cooperative
- Superteam Africa chapter application
- Grant applications: IFAD, World Bank, Solana Foundation
- Aadios Systems platform launch

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***AIM Protocol V2.1 — Production foundation complete. Built from Sierra Leone.***

Amjad Kamara | Aadios Systems (SL) Ltd. | <https://aim-protocol.vercel.app>  
*Built from Sierra Leone*